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Application of Euler Deconvolution to Magnetic Data for Mineral Exploration in the Arabian Shield

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Abstract

Euler Deconvolution is a widely utilized depth determination method that showed effectiveness characterizing geological model consisted on structures such as dykes, sills, faults and other simplified geometries showing lateral changes in magnetic susceptibility. The method's least-squares windowed approach renders its solutions more robust in comparison to similar methodologies. However, the price to pay for this robustness is the well-known scattering from the inversion process, commonly referred to in literature as "sprayed solutions". In this study, we apply this to synthetic data to examine the scattered behavior, as well as a traditional filter that relies on the expertise of the interpreter. To address this challenge, we propose an unsupervised machine learning algorithm for the generated solutions, facilitating a precise estimation of the locations of underlying magnetic sources.

A case study in the Arabian Shield was utilized for this approach to mineralized deposits. A profile, averaging 2 km in length, was strategically aligned perpendicular to the strike direction of the surface manifestations of the mineral deposits to optimize the detection of subsurface structures. The analysis of magnetic anomalies reveals a complex geological setup on the western side of the mineralized outcrop. Most of these anomalies are linked to sub-vertical sulfide bodies located near the surface. A qualitative assessment of the positive magnetic anomalies indicates the presence of magnetite-rich intrusions, possibly connected to porphyry systems. Conversely, negative anomalies detected over individual sulfide deposits and a broader magnetic low zone suggest influences beyond simply mound-related anomalies. The raw standard Euler method yields scattered results, making it challenging to determine the extent of geological body clustering from the magnetic anomaly data. However, applying a k-means clustering algorithm to these scattered outputs produced a reasonable estimate of the number of geological bodies responsible for those anomalies.

Overall, the results from the Euler method suggest that mineralized deposits are associated with linear features located at depths ranging from the surface to 10 meters. This study illustrates that combining the Euler deconvolution method with machine learning-driven techniques benefits mineral exploration due to its cost-effectiveness and rapid execution. This represents a significant advancement in mining exploration, particularly in identifying and characterizing subsurface mineral deposits during the greenfield stage, especially in the Arabian Shield and similar geological environments.

Introduction

In the early years, magnetic surveys were conducted by individuals who carefully moved from one point to another, recording the values they observed. As time progressed, vehicles became an option for covering wider areas. These vehicles were equipped with built-in magnetometers, allowing semi-automatic readings at intervals of less than one second. Today, airborne magnetic surveys—using fixed-wing aircraft, helicopters, or drones—are popular for collecting magnetic data, offering an impressive range, autonomy, rapid data gathering, and superior positional accuracy through GPS, marking a significant improvement over previous methods. This capability is particularly beneficial during the initial exploration phases when identifying target areas for further study is necessary. While airborne surveys excel in regional exploration, the detailed resolution data gathered from localized ground surveys can complement them effectively.

Thanks to recent advancements in instrumentation, it is now possible to map the entire crust more comprehensively, from strong magnetic basements at regional levels to weak magnetic sedimentary contacts at local levels. Additionally, when we combine magnetic modeling with additional geophysical data, we gain remarkable insights into the structures of interest, such as igneous and metamorphic basements in oil and gas exploration and ore bodies in mineral exploration. Moreover, advancements in computational performance have positioned Machine Learning as an attractive opportunity to tackle complex problems where conventional approaches fall short.

The Arabian Shield in Saudi Arabia is known for its mineral resources, including gold, copper, zinc, and other critical minerals for the energy transition. This study employs the Euler deconvolution technique on ground magnetic data within the Arabian Shield to assess its effectiveness in estimating the depth to magnetic sources related to mineral resources. The objective of this research is to support ongoing exploration efforts in the Arabian Shield region by applying the Euler deconvolution method and Machine Learning techniques to identify potential mineral exploration targets.

Theoretical Framework

Magnetic field gradients can either be calculated (Nabighian, 1972) or measured directly (Hood, 1965). Horizontal gradients are calculated from consecutive observed data points measured in a semi-horizontal plane. In contrast, the vertical gradient primarily relies on indirect calculations rather than direct measurements, employing Fourier transforms in the frequency domain (Nabighian, 1984). Amplitude derivatives are frequently used to delineate the boundaries of magnetic sources. However, their reliance on the induced magnetization poses a challenge in distinguishing minor magnetic anomalies when more significant and dominant anomalies are present. Therefore, the magnitude of the anomaly is influenced by the susceptibility contrast between the source and the surrounding geology.

Data Enhancement

After data correction, data enhancement becomes the next preliminary step to identify and map specific aspects of the spectral content—attributes and/or derivatives—contained in the anomaly that will help define the subsurface structures. These features can be correlated with other geophysical and geological data to provide insights for a better understanding of both the geology and the geometry of the structure. In simpler terms, it reveals "hidden" information by applying mathematical operators, such as horizontal and vertical

derivatives, to the magnetic anomaly, which is dominated by the response of the body's upper surface and edges (Fairhead, 2016).

Analytic Signal. The Analytics Signal, referred to as AS or Total Gradient, constitutes an amplitude derivative made up of the Pythagorean sum of the horizontal and vertical gradients of the magnetic field (Nabighian, 1972). The two-dimensional simplified case is expressed in the x-direction as follows:

$$AS(x) = \sqrt{\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial z}\right)^2} \quad (1)$$

The AS values remain positive, representing the Pythagorean sum of horizontal and vertical components. Additionally, its maxima are situated at the boundaries of magnetic bodies, with the amplitude proportionally linked to magnetization and susceptibility (Roest et al., 1992). The values of the analytic signal are not influenced by inclination and therefore remain unaffected by remanence.

Euler Deconvolution Method

Several approaches were developed to determine depths to magnetic sources based on the relationship between Euler's equation and magnetic source anomalies. However, the first workable application of this relationship, known as EULDPH (Thompson, 1982), focuses on 2-D magnetic profile data. Euler's homogeneity equation for profile magnetic data can be summarized as follows:

$$(x - x_0) \frac{\partial T}{\partial x} + (z - z_0) \frac{\partial T}{\partial z} = N(B - T) \quad (2)$$

Where x and z are the observation points in the x-axis and z-axis, x_0 and z_0 are the source's coordinates. T is the total magnetic intensity and B a background value. Mathematically, N is related to Euler's equation, defined for convenience as the negative value of the degree of homogeneity which depends on the type of field's anomaly, and can be nonunique for a given field element (Stavrev & Reid, 2007). Furthermore, N is interpreted physically as the attenuation rate of the anomaly field over distance, and geophysically as a structural index or SI. Several studies were conducted on this topic to define the source's geometry (Hood, 1965; Reid & Thurston, 2014). Obtaining SI values for structures such as contact ($N=0$), dyke ($N=1$), cylinder ($N=2$) and sphere ($N=3$).

Euler method showed effectiveness characterizing geological model consisted on structures such as dykes, sills, faults and other simplified geometries showing lateral changes in magnetic susceptibility (Reid et al., 2014). For more complex structures, the extended Euler method shows better-constrained solutions (Mushayandebvu et al., 2001). Compared to more recent depth determination techniques such as SPI (Thurston & Smith, 1997) and AN-EUL (Salem & Ravat, 2003), the Euler Deconvolution method showed robust solutions since it relies on least-squares inversion with moving windows using lower-order derivatives. This approach is helpful in practical cases where the noise level is reasonable, especially common in the oil&gas (Lopes de Castro, 2011) and mining (Cooper, 2024).

Solution filtering criteria. Traditionally, Euler's scattering is addressed by filtering the solutions using criteria that rely on statistics and spatial consensus. Most conventional criteria employ simple operations, ranging from rejecting solutions that fall outside the inversion's window width to applying low- and high-pass filters to constrain the frequency content, and concluding with an analysis of solutions using graphical and clustering techniques (FitzGerald et al., 2004).

Thompson's criteria

On the other hand, some criteria depend on the uncertainty of the data. For example, given a standard magnetic dataset, a depth estimate z_0 -corresponding to a given structural index N and window size—is accepted as a solution for plotting only if it satisfies the following condition (Thompson, 1982).

$$100 * \frac{N\sigma_z}{z_0} \leq TOL \quad (3)$$

While higher N often better represents most geological structures, lower indices typically correlate more directly with the depth to the top of the structure. The tolerance criterion " TOL " is empirically determined and must account for the depth of the estimate, as deeper z_0 inherently exhibits greater uncertainty values σ_z . Hence, most of the responsibility relies on the interpreter's expertise in choosing the proper parameters based on the acquisition setup and their geological knowledge of the area.

Machine Learning Clustering

The K-Means algorithm separates a dataset into K distinct clusters by minimizing the within-cluster sum of squares, also referred to as the inertia. Each cluster is defined by a centroid, calculated as the arithmetic mean of the observations assigned to that group. Although the cluster's centroids reside within the same feature space as the input data, they are not necessarily observation points from the dataset. The method assumes that clusters have approximately equal variance and requires the number of clusters, K , to be predefined. Due to its computational efficiency and scalability, K-Means has found widespread use across diverse scientific and engineering disciplines. Its primary objective is to optimize the distribution of data points among clusters to minimize intra-group variability and maximize inter-class variability (Pedregosa et al., 2011).

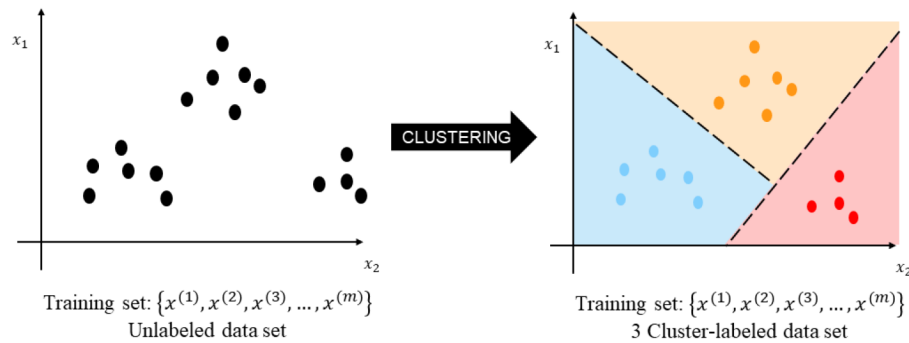


Figure 1—Clustering process from the unlabeled data set (black), identifying three main clusters (blue, orange, and red).

Giving an unlabeled training set consisted on " m " data $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$. The optimization objective relies on minimizing the cost function J as follows:

$$\min J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \min \|x^{(i)} - \mu_{c(i)}\|^2 \quad (4)$$

Being, $c^{(i)}$ index of K -ith cluster to which data point $x^{(i)}$ is currently assigned, $\mu_c(i)$ the cluster centroid of K -ith cluster to which example $x^{(i)}$ has been assigned. The right side of equation 3 consists of the inertia minimization process, which measures clusters' internal coherence, minimizing the within-cluster sum of squares, ensuring compact and well-separated clusters. One approach for choosing a suitable number of clusters is the Elbow method, which plots the inertia function versus K , where the best value is usually represented as an "elbow". The main objective is to reduce the sum of squared distances of points to their cluster center, also called inertia, by iteratively assigning point $c^{(i)}$ to the nearest centroid and updating $\mu_c(i)$ to the mean of their assigned points, resulting in K number of clusters where points are tightly grouped around their centroids ensuring a high intra-group similarity and low inter-class similarity using the minimum number of clusters.

The process begins with selecting K initial centroids, which may be chosen randomly or via a more sophisticated strategy such as K-Means++ to improve convergence properties (Arthur & Vassilvitskii, 2007). Each observation in the dataset is assigned to the nearest centroid using a selected distance metric, typically the Euclidean distance. This step divides the dataset into K provisional clusters. A new centroid is computed as the arithmetic mean of all points currently assigned to each cluster. These centroids serve as updated cluster centers for the next iteration. The algorithm then assesses whether the centroids

have changed significantly relative to the previous iteration. If the centroids remain stable, the algorithm converges, and the iteration terminates. If not, the assignment and update steps are repeated. Finally, once convergence is achieved, the algorithm outputs the final cluster memberships and the positions of the centroids (Figure 2).

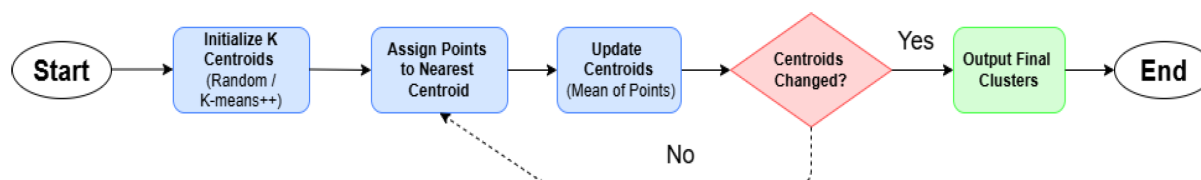


Figure 2—Basic workflow for the K-means++ clustering algorithm.

Synthetic Data

First, we tested the Euler Deconvolution method using synthetic datasets of theoretical and simplified structures to identify the main gaps in Euler's depth solutions and filtering criteria. One theoretical structure is a thin vertical dyke located 6 meters below the surface, extending to semi-infinite depth with induced magnetization. The magnetic anomaly is calculated along a 100 m profile using an induced magnetic field with a declination of 0 degrees and an inclination of 60 degrees. The second model consists of two vertical prisms: the western prism is located at a depth of 4 km, while the eastern prism is at 8 km. Both exhibit positive magnetization contrast and extend to semi-infinite depth. The magnetic anomaly is measured from a 300 m west-east profile, inducing a magnetic field with a declination of 0 degrees and an inclination of 90 degrees.

Thompson's criteria

The raw solutions from noise-free synthetic data exhibit Euler's inherent spraying resulting from the windowed scheme. This behavior is managed by applying Thompson's tolerance threshold of 20%, which is sufficient to mitigate the spraying behavior, leading to clustered solutions at the top edge of the dyke. However, for the two-prisms model, this threshold was not accurate; reducing the threshold to 1% still shows scattering and high uncertainties, particularly for the deeper structure due to interference (Figure 3a and Figure 3b).

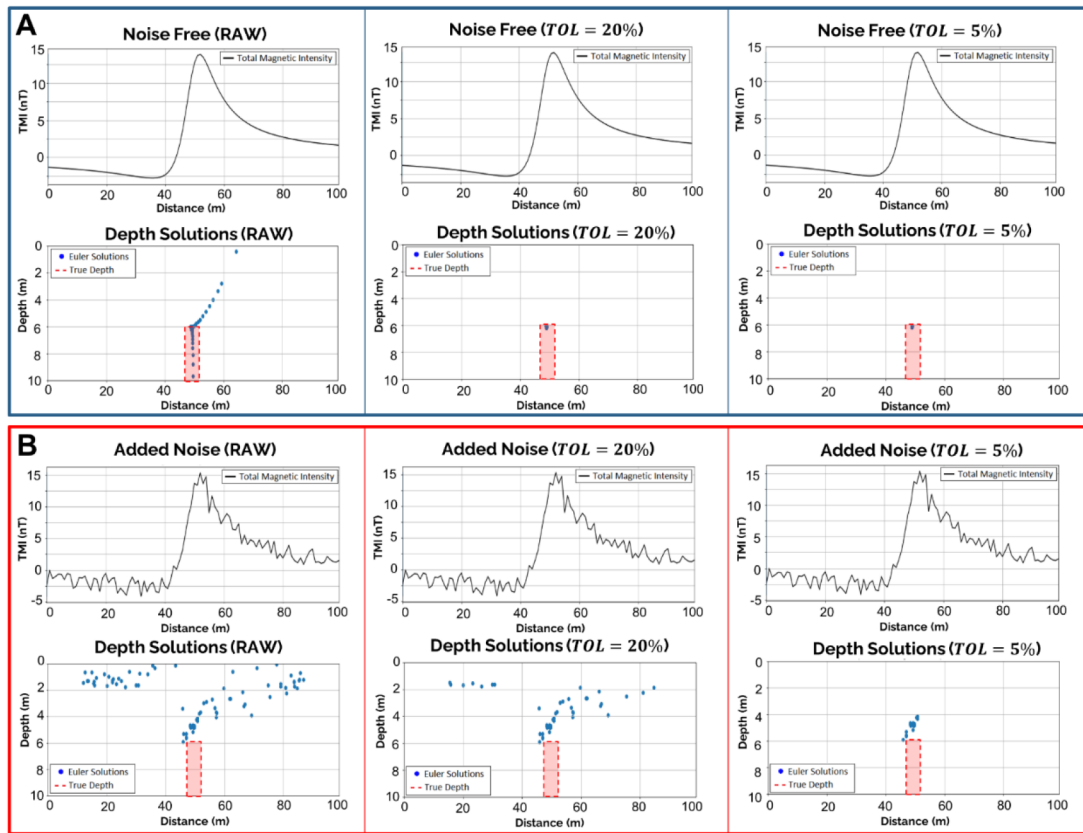


Figure 3—Euler solutions for a magnetic anomaly from a thin vertical dyke with: (A) Noise-free without filtering, Thompson's criteria with TOL=20% and Thompson's criteria with TOL=5%; (B) Added noise without filtering, Thompson's criteria with TOL=20% and TOL=5%.

After applying random noise to the previous data, the anomalies were affected by zero-mean oscillations (Figure 3b and Figure 4b). This is evident at the profile extremes, where the anomaly is relatively small. However, these oscillations are even more pronounced for the gradients obtained from the total field anomaly, as the derivative operator amplifies the high-frequency content of the noise. The result of this amplification is the underestimation of depths and the formation of shallow artifacts along the entire profile. For the dyke model, a tolerance threshold of 20% led to a partial filtering of these shallow solutions. However, deeper and closer solutions to the structure remain unfiltered since a lower uncertainty threshold is necessary, requiring partial depth solutions with a tolerance threshold of 5% between 4 and 6 meters.

Regarding the two-prism model, the solutions originate from surface depths as the inversion window approaches the shallow prism; they cluster at the top-edge boundary along the x-axis. However, the depth solutions along the z-axis vary widely from 4 to 6 meters, with outliers ranging from 8 to 10 meters. After applying Thompson's criteria, lateral resolution has been noted to improve. Nonetheless, even with the stricter threshold, the criteria still struggle to filter the depth solutions located at the source boundaries, resulting in the highest uncertainties in the deeper structure. Similar to the dyke model, the previous data is contaminated with zero-mean random noise. After applying a standard tolerance threshold of 20%, the surface solutions from high frequencies at the profile's edge were successfully filtered. However, no filtered solutions were found within the structure's domains. Finally, a stricter threshold of 1% was implemented.

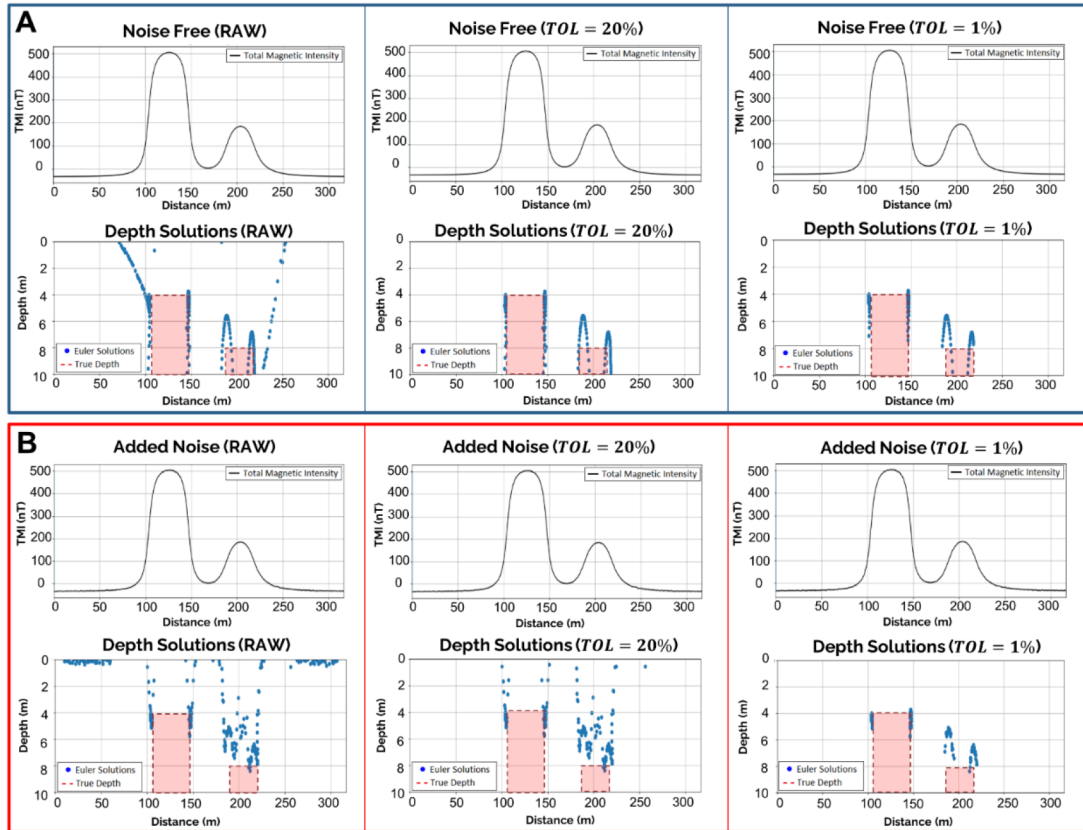


Figure 4—Euler solutions for a magnetic anomaly from a thin vertical dyke with: (A) Noise-free without filtering, Thompson's criteria with TOL=20% and Thompson's criteria with TOL=5%; (B) Added noise without filtering, Thompson's criteria with TOL=20% and TOL=1%.

ML-driven Technique

From the noise-free dyke, the cost function indicates two possible setups with two minimums at $K=1$ and $K=3$ (Figure 5a). However, the elbow in the cost function corresponds to a local minimum, leading to spurious results from solutions that cluster shallower; these solutions are related to the intrinsic spraying from the windowed technique, where high interclass variability is achieved. However, this is not feasible from a geological sense. For the noisy dyke, the results are similar, with minimum inertia at K values of 1 and 4 (Figure 5b). In this case, the noise increases the inertia values, and the elbow is located at a higher K , suggesting a value of 4. However, this approach fails three out of four times in identifying the top edge source. Again, the best solution is represented as a single cluster ($K=1$).

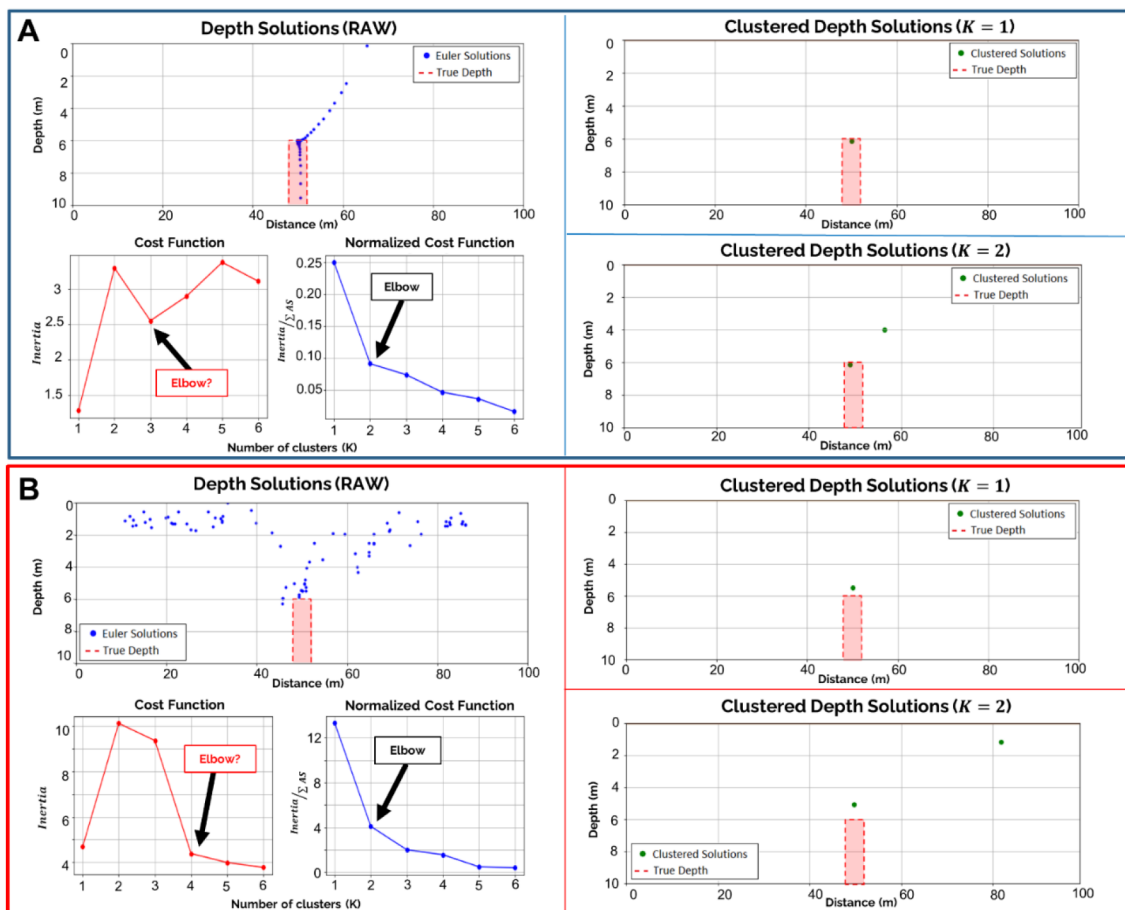


Figure 5—Euler solutions from two vertical prisms with: (A) Noise-free solutions and clustering cost function (left) and K-means+ clustered solutions for K=2 and K=5 (right); (B) Added noise solutions and clustering cost function (left), and K-means++ clustered solutions (right).

Regarding the two prisms, K-means clustering demonstrated exceptional results for the noise-free data, accurately identifying three out of four top edge structures. The solution with the highest uncertainty lies between both structures and serves as a clear indicator of interference noise inherent in the Euler solutions (Figure 6a). Although $K=4$ does not appear as a clear "elbow," this is because $K=5$ yields a local minimum, possibly due to poor initialization, suggesting a higher value. However, it is noted that this value resulted in an underestimation of the depths due to shallow artifact clustering in the solutions. Once we added noise to the data, the need for higher classes became evident as the inertia values increased compared to the previous mode. The elbow located at $K=5$ shows two solutions falling on the top edges of the shallow prism. Although it fails with the deep structure, the lateral solutions x_0 show promising results, with an underestimation of depths related to the high-frequency noises.

Finally, compared to the other filtering criteria, we observe an improvement in the solutions, especially for the shallower and stronger anomalies. However, this approach still struggles to effectively cluster the deeper solutions when using the elbow method to select an appropriate number of clusters. Furthermore, choosing a large number of clusters could result in artifacts (Figure 6b).

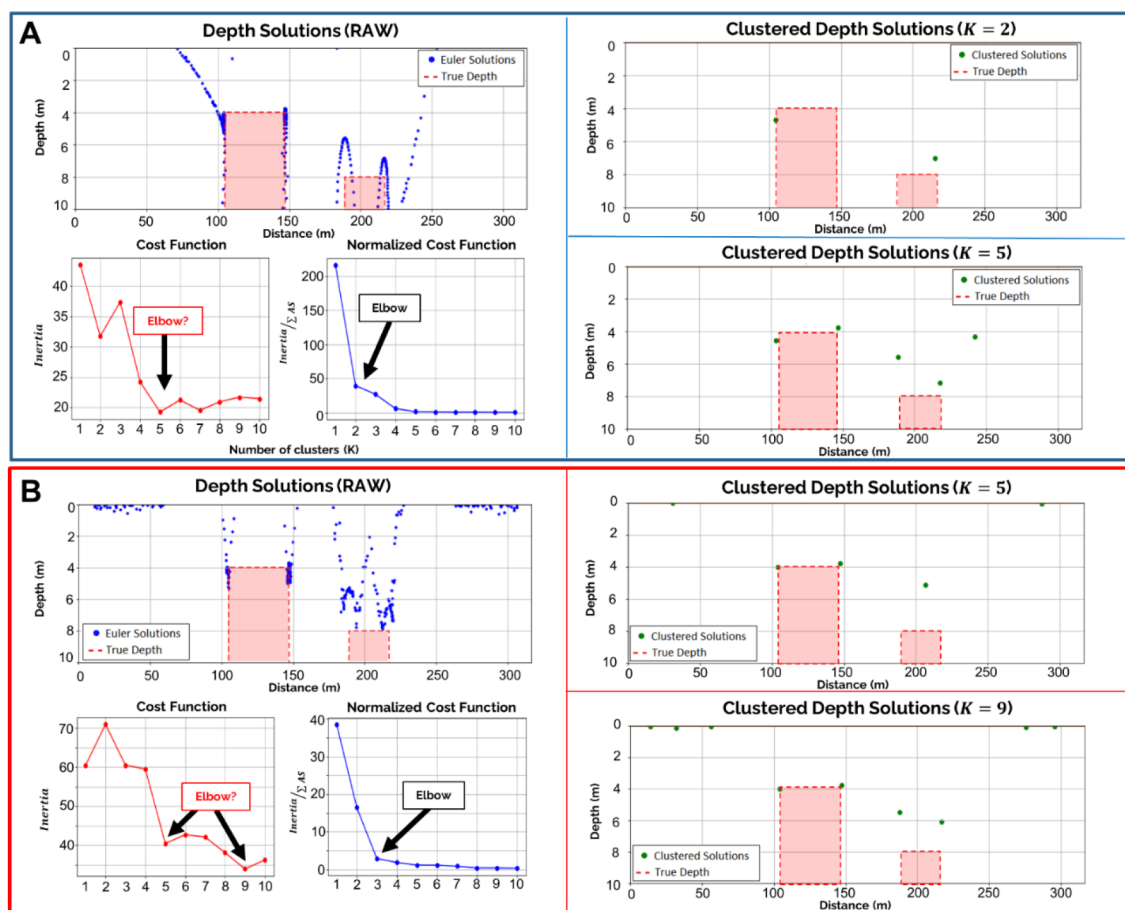


Figure 6—Euler solutions from two vertical prisms with: (A) Noise-free solutions and clustering cost function (left) and K-means+ clustered solutions for K=2 and K=5 (right); (B) Added noise solutions and clustering cost function (left), and K-means++ clustered solutions (right).

Field Data – Case Study

The study area encompasses the Hawiyah prospect in the Wadi Bidah Mineral District (WBMD), within the Asir Terrane of the Arabian Shield, located approximately 100 km east of Taif city, Saudi Arabia (Figure 7a). Recent initiatives in the Kingdom have made this district attractive for investment, having opened several mining concessions to evaluate its potential. Consequently, exploration in this mineral district is ongoing, with most projects still in the Greenfield stage (Workman et al., 2016). Most structures within the WBMD trend north-south (Figure 7b). Its extension includes 1400 km² of mineral deposits, comprising over 30 polymetallic Volcanogenic Massive Sulphides occurrences (Volesky et al., 2017). Locally, the alterations are expressed on the surface as north-south trending gossans (Figure 7d).

For the purpose of this paper, a single profile is presented as a case study from the Hawiyah Prospect in the Arabian Shield, illustrating the proposed ML-driven unsupervised clustering approach. The magnetic signatures exhibit anomalies in the center and western parts of the mineralized outcrop, suggesting a complex geological environment. Additionally, the dominant negative behavior of the total magnetic intensity and the asymmetry from the reduced-to-pole anomalies indicate remanent magnetization along the structure.

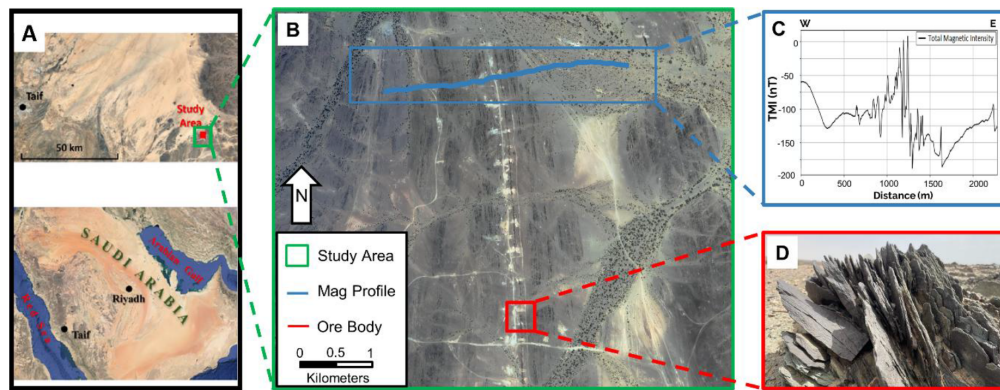


Figure 7—(A) Location of the study area; (B) Satellite imagery of the Hawiyah Prospect; (C) TMI anomaly from magnetic profile; (D) Surface manifestations as iron-rich gossans.

Regarding the Euler solutions, we employed a window of 10 meters for the deconvolution method, resulting in an overdetermined system that enhances both inversion robustness and lateral resolution. The Euler scattered solutions consisted of over 5000 solutions ranging from the surface to a depth of 30 meters; the deeper and more dispersed solutions originated from the edges of the profile, while the less scattered and denser solutions came from the center, ranging from 15 meters to the surface (Figure 8a). The surface solutions were considered spurious and attributed to high-frequency noise. An analytical signal window of the same length as Euler was employed, filtering out more than half of the solutions using the analytic signal peaks (Figure 8b). After applying the K-means clustering approach within the AS window, we present four possible clustering setups for the proposed depth solutions, all ranging from 1 to 10 meters and mostly located in the middle of the profile (Figure 8d).

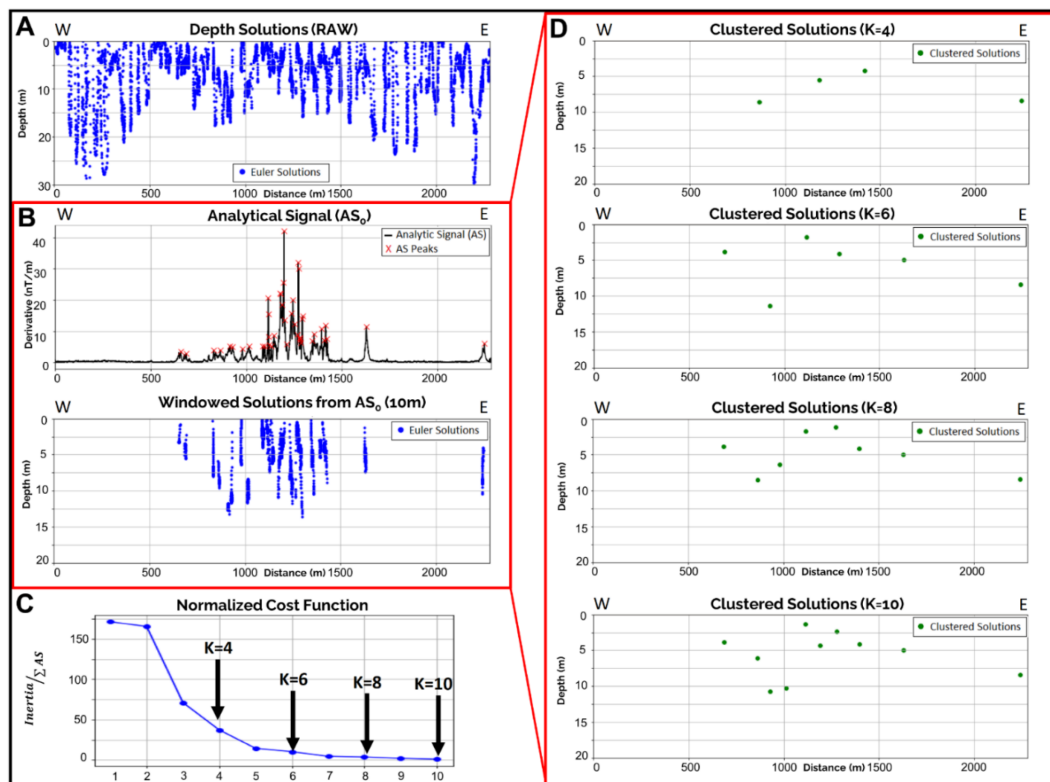


Figure 8—(A) Raw Euler solutions from TMI anomaly; (B) Analytical Signal and windowed solution from its peaks; (C) Cost function suggesting number of clusters; (D) Clustered solutions for the suggested classes.

As expected, the cost function during the clustering process based on field data shows a higher number of classes than those demonstrated by the synthetic models. This suggests a scenario involving noise originating from both cultural or magnetic debris and geological interference. Since we employ the elbow values to guide the clustering, it is observed that there is a necessity to increase the number of clusters beyond the "elbow" to cover all the analytical signal peaks along the profiles, as the majority of them are source indicators.

Discussion and conclusions

Although the solutions are closer together, a spread persists. As we have assumed, this occurs even with ideal sources and properly defined parameters. Furthermore, the spread is independent of the shape of the source. Instead, it relies on Euler's intrinsic moving-window scheme and the selection criteria derived from the proposed filtering techniques. However, it is noted that viable solutions from all models tend to cluster near the upper edges of the structure.

Two types of noise predominantly influence the solutions derived from synthetic data. Firstly, there is random noise from the acquisition stage, particularly due to the instrument or cultural background. Secondly, interference noise arises from adjacent sources, where the presence of a closer, shallower source distorts the solutions from deeper structures. In summary, the assumptions regarding appropriate parameters for Euler (window length, structural index)—all derived from the previously discussed filtering criteria—have a sound theoretical basis. However, they may fail in practice, as both types of noise (inherent in the data and the geology) contribute to the scattering of solutions. It is crucial to filter the surface artifacts primarily generated by high-frequency noise at the profile margins, where the magnetic anomalies from true sources are absent.

Regarding the case study, raw field data solutions derived from the standard Euler method are highly scattered and cannot directly determine the number of geological bodies clustered from the observed magnetic anomaly data, relying heavily on the interpreter's expertise in selecting appropriate tolerance parameters. However, once the k-means unsupervised clustering algorithm is applied to the scattered solutions, we obtain a reasonable number of geologic bodies that produced these solutions, mainly located at the top edge of the bodies. In this case, the preliminary results suggest that the mineralized deposits are associated with linear features, with depth solutions ranging from the surface to 10 meters.

It is essential to acknowledge that although this methodology is not fully automated, it enhances the standard semi-automatic Euler deconvolution method. The proposed machine learning-driven criterion relies more on the inherent characteristics of the data and involves fewer suggestive parameters, thereby reducing potential interpreter bias. Nevertheless, determining the appropriate number of geological features within the clusters presents a challenge, as the "elbow" is not clearly visible, making the choice ambiguous. Consequently, selecting an appropriate value for the number of classes is a compromise between achieving good interpretability from a limited number of clusters, which typically oversimplifies the results, and attaining complex interpretability from lower inertias with a greater number of clusters, which generally risks overfitting to noise. One potential solution is to refine the existing Elbow Method, explore alternative techniques for determining the optimal number of clusters (such as the Silhouette Method), or directly avoid the problem with a different clustering algorithm (e.g., DBSCAN).

Finally, a significant advantage of combining ML-driven criteria with Euler Deconvolution is its ability to handle intricate, non-linear relationships in geophysical data that traditional methods might overlook. This integration creates new opportunities for interdisciplinary studies in geophysics and AI. The distinction between ML and deterministic techniques, such as Euler Deconvolution, is evident in their data interpretations, adaptability, and handling of uncertainty.

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